Incident Response Lab Guide

One hands-on lab per video. The labs build on each other: the environment from Lab 3, the plan from Lab 6 and the case file you start in Lab 4 are used through to the end-to-end incident in Lab 29.

How to use this guide

Do each lab after its video. Every lab lists an objective, the earlier labs it depends on, the steps, what you hand in, and a completion check. Don't treat a lab as done until it passes the check.

Times are estimates for a solo learner with the lab already built. Team exercises (Labs 11, 23, 29) run better with two or more people; working solo, play each role in turn and write down every handoff.

Lab rules

1. **Isolate.** The lab has no route to your home, corporate or production networks. Check this before every lab that involves attack tooling.
2. **Only systems you own.** Never run attack or acquisition tools against systems you don't own or have written permission to test.
3. **Malware stays in the analysis environment.** Samples are stored compressed and password-protected, are never opened on the host, and are never run with internet access.
4. **Snapshot first.** Take a named snapshot before each lab and record it in the case log.
5. **UTC everywhere.** Every timestamp in your notes, timelines and custody records is in UTC.
6. **Write it down as you go.** If it isn't in the case log, it didn't happen.

The case file

You start a case file in Lab 3 and use it for every incident after that. Keep the same layout for every case:

FOLDER	CONTENTS
`notes/`	Running case log: UTC time, who, what was done, why, and the result
`evidence/`	Acquired artifacts, write-protected, never modified
`hashes/`	SHA-256 for each artifact at acquisition and after every transfer
`custody/`	Chain-of-custody forms, one per evidence item
`timeline/`	The master timeline (time, source, host, event, interpretation, confidence)
`reports/`	Deliverables: plans, briefs, reviews, memos

Lab map

LAB	TITLE	TIME
01	Separate the attack from the response	60 min
02	Walk one incident through the lifecycle	45 min
03	Build the incident response lab	3–4 h
04	Acquire evidence with a chain of custody	90 min
05	Write the response approach	60 min
06	Write the incident response plan	3 h
07	Design the incident response team	90 min
08	Write a playbook	2 h
09	Set up centralized logging and build a detection	3 h
10	Assemble and test the response toolkit	2–3 h
11	Run a tabletop exercise	2 h
12	Triage a set of alerts	90 min
13	Scope before you contain	2 h
14	Reconstruct the attack timeline	3 h
15	Do disk forensics on a compromised host	3 h

LAB	TITLE	TIME
16	Capture and analyze memory	2–3 h
17	Investigate the network and the cloud	3 h
18	Analyze malware as a responder	2–3 h
19	Contain the incident in one coordinated move	2 h
20	Eradicate the attacker	2–3 h
21	Recover and restore operations	2–3 h
22	Draft the incident communications	2 h
23	Coordinate a major incident	4 h, team
24	Run a post-incident review	2 h
25	Handle ransomware from detection to recovery	4 h
26	Investigate an account takeover	2–3 h
27	Determine whether a breach occurred	2–3 h
28	Respond to a cloud incident under shared responsibility	3 h
29	Handle a full incident end to end	1 day
30	Analyze a real response failure	90 min
31	Assess program maturity and build a roadmap	2 h
32	Plan your skills and next steps	60 min

PART I · LABS 01-05

Foundations

Learn the vocabulary, the lifecycle and the habits of a responder, and build the lab every later exercise runs in.

LAB 01 VIDEO 1 60 MIN BUILDS ON: —

Separate the attack from the response

Objective. Take a public breach, sort the attacker's actions from the defenders' actions, and judge where the response helped or hurt.

STEPS

01 Pick a breach that has a detailed published timeline: a post-incident report, regulator finding or court filing.

02 Build a two-column timeline: attacker actions on the left, the organization's actions on the right, both in UTC.

03 Mark when each threshold was crossed: event, alert, incident, breach. Note any point where the organization confused two of them.

04 Label each response action as helped, hurt or neutral, with one sentence of reasoning.

05 Identify the one response decision that changed the outcome most.

DELIVERABLES

- Two-column timeline with sources
- One-page assessment of the response

COMPLETION CHECK

Every timeline entry cites a source, and you can say exactly when the activity became an incident and when it became a breach.

LAB 02 VIDEO 2 45 MIN BUILDS ON: 01

Walk one incident through the lifecycle

Objective. Take one simple incident through all six lifecycle stages on paper and assign an owner to each stage.

STEPS

01 Write a three-sentence scenario. Suggested: a staff member reports entering their password into a login page after clicking an email link.

02 For each stage (preparation, detection and analysis, containment, eradication, recovery, lessons learned), write what happens, which decision ends the stage, and who owns it.

03 Add the four parallel tracks as rows: technical, communication, legal and regulatory, business continuity. Fill in each track for each stage.

04 Mark at least one point where the incident loops back from containment to detection and analysis.

05 Write down what lessons learned would feed back into preparation.

DELIVERABLES

- Stage-by-track grid (6 × 4)

COMPLETION CHECK

Every cell has an owner, and the loop back is drawn explicitly.

LAB 03 VIDEO 3 3-4 H BUILDS ON: —

Build the incident response lab

Objective. Set up an isolated victim environment and a separate analysis environment, and confirm you can safely move evidence from one to the other.

Safety. Confirm the lab is isolated before installing any attack tooling.

STEPS

01 Create an isolated virtual network (host-only or internal switch) with no route to any real network.

02 Build the victim environment: an identity server or domain controller, two workstations and one Linux server, with test users and seeded "sensitive" data.

03 Build the analysis workstation on its own isolated segment. Install disk acquisition, memory analysis and log analysis tools (for example FTK Imager or dd, Volatility 3, and a log stack).

04 Create the case-file folder structure and make your first case-log entry.

05 Take a named baseline snapshot of every VM and record each in the log.

06 Test the transfer path: acquire a file on the victim, hash it, move it through a one-way path (a read-only share or an attached virtual disk), and hash it again.

DELIVERABLES

- Network diagram
- Snapshot inventory
- First case-log entry

COMPLETION CHECK

The hashes match, the analysis machine can reach neither the internet nor the victim except through the controlled path, and you can revert to baseline in under five minutes.

LAB 04 VIDEO 4 90 MIN BUILDS ON: 03

Acquire evidence with a chain of custody

Objective. Acquire evidence from a lab system in a forensically sound way and keep a complete chain of custody for it.

STEPS

01 Open a case. Start a custody form for each item: description, collector, UTC date and time, source location and hash.

02 Capture in order of volatility: memory first, then running processes and network connections, then the disk image.

03 Hash each artifact (SHA-256) when you acquire it and store it write-protected.

04 Move it to the analysis environment. Log every handoff: time, from, to and purpose.

05 Check the hashes after the transfer and again after your first analysis session.

06 Write one paragraph on what you did to the live system and what your actions changed.

DELIVERABLES

- Custody forms
- Hash log
- Acquisition notes

COMPLETION CHECK

A reviewer can trace each artifact from its source to where it is now, with no gaps.

LAB 05 VIDEO 5 60 MIN BUILDS ON: 02

Write the response approach

Objective. Plan the order of your actions for a described incident so that you contain the damage without destroying evidence.

STEPS

01 Scenario: EDR flags an unsigned binary beaconing from a finance workstation, and the same user's account logged into a file server at 03:12 UTC.

02 List what is known, what is suspected and what is unknown.

03 Write the scoping questions that must be answered before you act.

04 Put your first ten actions in order. Mark each one as preserving evidence, containing, or risking tipping off the attacker.

05 Name one action you will deliberately hold back for now, and say why.

06 Define the trigger for moving from observing to containing.

DELIVERABLES

- Ordered action plan with reasoning

COMPLETION CHECK

No eradication step comes before containment, and memory is captured before any reboot or isolation that would lose it.

PART II · LABS 06–11

Preparation

Incidents are won or lost before they start. These labs produce the plan, team, playbooks, visibility, toolkit and rehearsal that the rest of the course relies on.

LAB 06 VIDEO 6 3 H BUILDS ON: 03, 05

Write the incident response plan

Objective. Write an incident response plan for your lab organization that a stressed team can actually follow.

STEPS

01 Define the organization: size, critical systems and regulated data.

02 Write the activation criteria: what counts as an incident, and who can declare one.

03 Define severity levels SEV1–SEV4 with impact, examples and a required response time for each.

04 List roles with a primary and a backup for each. Add a decision-authority table: who can disconnect a system, notify customers, contact law enforcement, and declare a major incident.

05 Write the escalation path and contact list, and name an out-of-band channel for when email and chat are compromised.

06 Store an offline copy and record where it lives.

DELIVERABLES

- Incident response plan, 10 pages or fewer

COMPLETION CHECK

Someone who hasn't read the plan can use it to classify three sample events and name who decides on isolation, within two minutes.

LAB 07 VIDEO 7 90 MIN BUILDS ON: 06

Design the incident response team

Objective. Design a team structure in which every role has a backup and no one is a single point of failure.

STEPS

01 Choose a team model (dedicated, virtual or hybrid) and explain why it fits.

02 Define the core roles (incident commander, technical lead, analysts, forensics, scribe) and the extended team (legal, communications, HR, executive, business owners).

03 Name a backup for every role. Check that no one person is the only holder of a critical skill or access.

04 Define coverage: the on-call rotation, the handover process and the maximum shift length.

05 Remove each person in turn and check that the structure still works.

DELIVERABLES

- Team chart
- RACI matrix
- Coverage and rotation plan

COMPLETION CHECK

Every role has a backup who is not on the same shift.

LAB 08 VIDEO 8 2 H BUILDS ON: 06, 07

Write a playbook

Objective. Write a complete playbook for one incident type, with triggers, steps, decision points and escalation.

STEPS

01 Pick an incident type. Suggested: phishing where the user entered credentials.

02 Define the triggers that start the playbook.

03 Write the steps, grouped by lifecycle stage.

04 Add decision points with explicit criteria, and escalation criteria tied to the severity levels from Lab 06.

05 List every tool and access the playbook assumes, and confirm each one exists in your lab.

06 Mark which steps could be automated and which need human judgment.

DELIVERABLES

- Playbook
- One linked runbook for a technical step

COMPLETION CHECK

Walk the playbook through a simulated case. No step depends on a tool or access you don't have.

LAB 09 VIDEO 9 3 H BUILDS ON: 03

Set up centralized logging and build a detection

Objective. Set up centralized logging, write a detection for one attacker behavior, and prove it fires.

STEPS

01 Deploy endpoint telemetry (for example Sysmon on Windows, auditd on Linux) and forward it to a central log platform.

02 Collect authentication, process, network and DNS logs. Confirm time sync (NTP) across all hosts.

03 Set retention for each source and record it.

04 Pick one ATT&CK technique (for example T1053, Scheduled Task) and write the detection logic.

05 Simulate the behavior, then a harmless action that looks similar. Tune until the rule fires on the first and not the second.

06 Write down one coverage gap you found.

DELIVERABLES

- Logging inventory (source, key fields, retention)
- Detection rule with its ATT&CK mapping
- Test evidence

COMPLETION CHECK

The alert fires within your target time on the simulated attack and doesn't fire on the harmless lookalike.

LAB 10 VIDEO 10 2–3 H BUILDS ON: 03, 09

Assemble and test the response toolkit

Objective. Put your response toolkit together and prove it can detect, investigate and acquire evidence before you need it.

STEPS

01 List your tools by category: detection and alerting, endpoint investigation, forensic acquisition, log analysis, case management.

02 Deploy the endpoint agent to every victim host. Confirm it can query and isolate each one.

03 Test the acquisition tool on a known file. Check that the hashes match and the source timestamps are unchanged.

04 Build a portable responder kit: offline binaries, their hashes and documentation.

05 Write a one-page quick reference for each tool.

DELIVERABLES

- Toolkit matrix (tool, category, where deployed, date tested, owner)
- Quick-reference sheets

COMPLETION CHECK

Run detect, investigate and acquire end to end on one host using only the kit.

LAB 11 VIDEO 11 2 H BUILDS ON: 06, 07, 08

Run a tabletop exercise

Objective. Run a tabletop exercise on a realistic scenario and record every gap it reveals.

STEPS

01 Write a scenario with four to six injects. Suggested: a ransom note found on a file server at 17:00 on a Friday.

02 Assign roles from Lab 07, and appoint a facilitator and a note-taker.

03 Work through the injects with a time limit on each decision. Record each decision, who made it and when.

04 At each inject, ask: who decides, what does the plan say, and what is missing?

05 Hold a hot wash straight afterward.

06 Sort the gaps into plan, roles, communication and tooling.

DELIVERABLES

- Exercise report
- Gap register

COMPLETION CHECK

Every gap has an owner and points to the plan or playbook section that needs to change.

PART III · LABS 12-18

Detection and Analysis

Turn a signal into a decision, then understand the incident well enough to act: scope, timeline, and the disk, memory, network, cloud and malware evidence behind it.

LAB 12 VIDEO 12 90 MIN BUILDS ON: 09

Triage a set of alerts

Objective. Triage a mixed set of alerts, separate the noise from a real incident, and decide what to escalate.

STEPS

01 Generate a mixed set: routine admin activity, a failed brute-force attempt, a scanner, and one real attack chain (seed it with a simulation framework such as Atomic Red Team).

02 For each alert, answer: what fired, what it means, and how confident you are.

03 Classify each alert as nothing, something or an emergency.

04 Add context to the "something" alerts: user, host and history.

05 Decide on escalation using the severity criteria from Lab 06. Record how long each triage took.

DELIVERABLES

- Triage sheet with classification, confidence and time per alert

COMPLETION CHECK

The real chain is escalated, and no harmless alert is escalated without a written reason.

LAB 13 VIDEO 13 2 H BUILDS ON: 12

Scope before you contain

Objective. Work out the full extent of the compromise before planning any containment.

STEPS

01 Start from the escalated alert. List what is known, suspected and unknown.

02 Search every host for the indicators you have so far: hashes, IPs and accounts.

03 Identify the entry point, the affected systems and accounts, and the attacker's apparent objective.

04 Start the master timeline.

05 Write a stakeholder update that keeps facts and hypotheses clearly apart.

06 Don't contain anything yet. List the containment candidates for Lab 19.

DELIVERABLES

- Scope statement
- Affected-asset list with a confidence level for each

COMPLETION CHECK

Compare your scope with the scenario's answer key and record any foothold you missed.

LAB 14 VIDEO 14 3 H BUILDS ON: 13

Reconstruct the attack timeline

Objective. Merge several log sources into one ordered account of the attack.

STEPS

01 Export authentication, process, network, file and cloud logs.

02 Convert all timestamps to UTC. Record each source's original time zone and any clock skew.

03 Work forward from the initial access and backward from the latest activity.

04 Pivot: turn each new artifact (account, IP, hash) into your next query.

05 Combine everything into one table: time, source, host, event, interpretation, confidence.

06 Mark the gaps explicitly, and say what each gap could be hiding.

DELIVERABLES

- Master timeline
- One-page narrative

COMPLETION CHECK

Every row cites its log source, and another analyst can reproduce every interpretation.

LAB 15 VIDEO 15 3 H BUILDS ON: 04, 14

Do disk forensics on a compromised host

Objective. Find the attacker's tools, persistence and activity on a compromised system's disk.

STEPS

01 Work from the image acquired in Lab 04, never the live disk. Mount it read-only.

02 Extract evidence of execution: Prefetch, Amcache, ShimCache and shell history.

03 Look for persistence: Run keys, services, scheduled tasks, WMI subscriptions, cron, systemd units and startup folders.

04 Recover the attacker's tools and staging directories, including deleted files.

05 Compare with the baseline snapshot to separate normal system behavior from attacker activity.

06 Add your findings to the master timeline.

DELIVERABLES

- Artifact table
- Persistence list

COMPLETION CHECK

The image hash is unchanged after your analysis.

LAB 16 VIDEO 16 2-3 H BUILDS ON: 04, 15

Capture and analyze memory

Objective. Capture and analyze memory from a compromised system and find evidence that doesn't exist on disk.

STEPS

01 Capture memory from the running compromised VM before doing anything else (or suspend the VM and copy its memory file).

02 Hash the capture and move it to the analysis environment.

03 Compare process listings with process scans to find hidden processes. Look for odd parent-child relationships.

04 List network connections and scan for injected code.

05 Run a fileless technique (for example an in-memory PowerShell payload) and confirm it leaves nothing on disk.

06 Link your memory findings to the timeline from Labs 14 and 15.

DELIVERABLES

- Memory findings
- At least one finding proven to be absent from disk

COMPLETION CHECK

The memory-only finding is confirmed absent from the disk image.

LAB 17 VIDEO 17 3 H BUILDS ON: 14

Investigate the network and the cloud

Objective. Trace lateral movement and identify exfiltration across the network and a cloud account.

STEPS

01 Capture traffic at the lab switch (pcap, or Zeek logs).

02 Identify command and control: beacon interval, destination and protocol.

03 Trace lateral movement by matching authentication logs with SMB, RDP, WinRM and SSH flows from the entry host.

04 Measure the exfiltration: volume, destination and time window.

05 In a test cloud account, review the audit trail (for example CloudTrail or the Azure Activity Log) for the compromised identity's actions.

06 Write down what the provider's logs didn't show you.

DELIVERABLES

- Lateral-movement map
- Exfiltration determination with evidence

COMPLETION CHECK

Every hop on the map is backed by both an authentication record and a network record.

LAB 18 VIDEO 18 2–3 H BUILDS ON: 03, 13

Analyze malware as a responder

Objective. Analyze a malware sample and extract the indicators and persistence you need to scope and contain the incident.

Safety. Analysis environment only. Snapshot first. No network, or a simulated one (for example INetSim).

STEPS

01 Static triage: hashes, strings, imports and packer detection. Check reputation by hash only — never upload the sample.

02 Behavioral analysis: run it in the sandbox and record process, file, registry and network activity.

03 Extract the indicators of compromise and the persistence mechanism.

04 Search the whole environment for those indicators and find every infected host.

05 Decide whether deeper reverse engineering is needed. If it is, write the handoff note.

DELIVERABLES

- Machine-readable IOC list
- Capability summary

COMPLETION CHECK

Your search found every infected host in the answer key, and the analysis VM was reverted afterward.

PART IV · LABS 19–23

Containment, Eradication and Recovery

Stop the damage, remove the attacker completely, and bring the business back safely, while running communication and coordination alongside.

LAB 19 VIDEO 19 2 H BUILDS ON: 13–18

Contain the incident in one coordinated move

Objective. Plan and carry out a containment that cuts off all of the attacker's access at once.

STEPS

01 From your scoping, list every access path: hosts, accounts, sessions, tokens, C2 channels and persistence.

02 Choose a short-term and a long-term containment action for each path.

03 Capture any remaining volatile evidence first.

04 Plan a single containment window. Give every action an owner and a time.

05 Carry out all actions together, then confirm each one worked.

06 Watch for the attacker's reaction.

DELIVERABLES

- Containment plan
- Execution log

COMPLETION CHECK

No attacker channel survives, and C2 traffic stops on every host at the same time.

LAB 20 VIDEO 20 2–3 H BUILDS ON: 15–19

Eradicate the attacker

Objective. Remove every persistence mechanism, close the entry point, and prove the attacker is gone.

STEPS

01 Remove every persistence mechanism found in Labs 15–18, and record each removal.

02 Close the entry point: patch it, fix the configuration, or remove the exposure.

03 Rotate every credential the attacker could have touched, including service accounts (and krbtgt twice if the domain was compromised).

04 Decide for each host whether to rebuild or clean it, and write down why.

05 Verify: search for the indicators again, re-run your detections, and try the attacker's known ways back in.

DELIVERABLES

- Eradication checklist
- Rebuild-vs-clean decisions

COMPLETION CHECK

Every attempt to get back in fails, and the indicator search comes back clean.

LAB 21 VIDEO 21 2–3 H BUILDS ON: 20

Recover and restore operations

Objective. Restore the environment cleanly and watch for the attacker trying to return.

STEPS

01 Confirm the backups predate the compromise. Scan a test restore in isolation.

02 Restore in stages, in order of business priority.

03 Before reconnecting each system, check that it works, is intact and scans clean.

04 Turn on heightened monitoring for the attacker's indicators and techniques for a defined period.

05 Define the closure criteria. Declare the incident closed only when all of them are met.

DELIVERABLES

- Recovery log
- Closure memo

COMPLETION CHECK

A simulated attempt by the attacker to return is detected during the monitoring period.

LAB 22 VIDEO 22 2 H BUILDS ON: 04, 14

Draft the incident communications

Objective. Draft every communication a simulated breach needs.

STEPS

01 Internal team update: facts, actions taken and the time of the next update.

02 One-page executive brief: business impact, decisions needed and risk.

03 External holding statement: acknowledge the incident and say what is being done. No speculation.

04 Draft the regulatory notification, with its deadline calculated from the obligations covered in Lab 04.

05 Review every draft for over-disclosure and for anything not yet confirmed.

DELIVERABLES

- Internal update
- Executive brief
- Holding statement
- Regulatory notification draft

COMPLETION CHECK

Every factual claim matches a row in the timeline, and nothing unconfirmed is stated as fact.

LAB 23 VIDEO 23 4 H, TEAM BUILDS ON: 06, 07, 22

Coordinate a major incident

Objective. Run a major simulated incident with several workstreams and a shift handover.

STEPS

01 Declare a major incident. Set up command: the incident commander plus leads for the technical, communications, legal and business-continuity workstreams.

02 Set the battle rhythm: status update intervals and a decision log.

03 Run the workstreams in parallel against the injects.

04 Halfway through, hand over using a written template. The outgoing team leaves the room.

05 The incoming team continues using only the handover document.

06 In the debrief, list what context was lost.

DELIVERABLES

- Decision log
- Handover document
- List of lost context

COMPLETION CHECK

Within ten minutes, the incoming team can state the current scope, the open decisions and the next actions.

Lessons Learned

The stage that closes the loop. The incident is only worth what you learn from it and actually act on.

LAB 24 VIDEO 24 2 H BUILDS ON: ANY HANDLED INCIDENT

Run a post-incident review

Objective. Run a full post-incident review of a lab incident you handled and produce concrete, tracked improvement actions.

STEPS

01 Write down blameless ground rules before you start.

02 Reconstruct the full timeline, including the response actions.

03 Answer three questions: how did the attacker get in, why wasn't it caught sooner, and what slowed the response?

04 Keep failures of the response separate from failures of the defenses.

05 Turn each finding into an action with an owner, a due date and a measure of success. Link it to the plan, a playbook or a detection.

06 Track the actions to completion, and re-test one of them.

DELIVERABLES

- Post-incident review report
- Action tracker

COMPLETION CHECK

At least one action is implemented and verified before you close the lab.

Incident Types

The incident types that dominate today, and the ones that break the assumptions of traditional response.

LAB 25 VIDEO 25 4 H BUILDS ON: 09, 19-21

Handle ransomware from detection to recovery

Objective. Handle a ransomware incident from early detection through clean recovery from immutable backups.

Safety. Use a harmless encryption simulator. Never use real ransomware.

STEPS

01 Set up immutable or offline backups before you start.

02 Simulate the stages before encryption: initial access, discovery, credential theft, and staging and exfiltration.

03 Detect at least one of those stages.

04 Time the containment race: how long from detection to isolation.

05 Run the encryption simulator on a subset of hosts.

06 Recover from the immutable backup and confirm the restored systems are clean. Write down what the data theft means for extortion.

DELIVERABLES

- Detection record by stage
- Containment and recovery times

COMPLETION CHECK

The simulated attacker never touched the backups, and the restored systems are clean.

LAB 26 VIDEO 26 2–3 H BUILDS ON: 17, 19

Investigate an account takeover

Objective. Investigate a simulated account takeover, trace what the attacker did, and remove all their access, including persistent sessions.

STEPS

01 In a test tenant, compromise a test account with a simulated phished credential and a replayed session token.

02 As the attacker, create a mailbox rule, consent to an OAuth app, add an MFA method and set up forwarding.

03 Investigate using the sign-in and audit logs. Tell the attacker's actions apart from the user's by IP, device and user agent.

04 Contain: reset the password, revoke sessions and refresh tokens, and remove the added MFA methods, rules and apps.

05 Check for a second compromised account.

DELIVERABLES

- Action list, each attributed to the attacker or the user
- Identity containment checklist

COMPLETION CHECK

The replayed session no longer works after containment.

LAB 27 VIDEO 27 2–3 H BUILDS ON: 14, 17, 22

Determine whether a breach occurred

Objective. Investigate a simulated data breach, establish what data was exposed, and work out the notification obligations.

STEPS

01 Identify the data stores the attacker touched, using Labs 14 and 17.

02 Classify the data: personal, financial, health or credentials.

03 Determine whether data was accessed or exfiltrated, and cite the evidence for each.

04 Estimate the number of affected records and individuals in each jurisdiction.

05 Map the findings to notification obligations (for example GDPR's 72 hours, US state laws and sector rules), and mark where counsel makes the call.

06 Insider variant: write down how the investigation would change if the actor were an employee.

DELIVERABLES

- Breach determination memo

COMPLETION CHECK

Every claim of exfiltration has evidence behind it, and anything uncertain is flagged as uncertain.

LAB 28 VIDEO 28 3 H BUILDS ON: 17

Respond to a cloud incident under shared responsibility

Objective. Run a cloud incident while working within the evidence limits of the shared-responsibility model.

STEPS

01 In a sandbox cloud account, simulate a leaked access key being used to list and copy storage.

02 Confirm audit logging is on, and note which data-access events have to be enabled separately.

03 Investigate using only the provider's logs.

04 Contain: disable the key, revoke sessions and tighten the policies.

05 List the evidence you couldn't get and what you would ask the provider for.

06 Optional: a container incident. Capture the evidence before the pod is rescheduled.

DELIVERABLES

- Investigation summary
- Register of evidence limits

COMPLETION CHECK

You can name every action taken with the key, or explain exactly why you can't.

PART VII · LABS 29–32

Bringing It Together

One complete incident, a study of how responses fail, the program around the capability, and your path from here.

LAB 29 VIDEO 29 1 DAY BUILDS ON: ALL

Handle a full incident end to end

Objective. Handle a complete incident from detection through post-incident review, recording every decision.

STEPS

01 Reset the lab to baseline. Have a partner, or a script, run a scenario you haven't seen.

02 Work every lifecycle stage using your own plan, playbooks and toolkit.

03 Keep a decision log: the decision, the time, the information available and your reasoning.

04 Run the communication track alongside.

05 Finish with a post-incident review.

DELIVERABLES

- Complete case file

COMPLETION CHECK

A reviewer can reconstruct every decision from the case file alone.

LAB 30 VIDEO 30 90 MIN BUILDS ON: 01, 24

Analyze a real response failure

Objective. Analyze a documented real-world incident-response failure and identify which lifecycle stage it broke at, and why.

STEPS

01 Choose a documented failure: a regulator report, court filing or published review.

02 Map its events to the lifecycle stages.

03 Find the stage where it broke. Separate the damage that was unavoidable from the damage the response caused.

04 Identify the preparation that would have prevented it.

05 Compare it with the gaps you found in your own lab work.

DELIVERABLES

- Two-page failure analysis

COMPLETION CHECK

The breaking point is tied to a specific stage and backed by the source documents.

LAB 31 VIDEO 31 2 H BUILDS ON: 06–11, 24

Assess program maturity and build a roadmap

Objective. Assess an incident response program against a maturity model and build a roadmap to improve it, with metrics to measure progress.

STEPS

01 Choose a maturity model (for example NIST CSF tiers or SIM3).

02 Score each program element: plan, team, tooling, playbooks, exercises and metrics.

03 Calculate baseline time to detect, contain and recover from your lab incidents.

04 Name the vanity metrics to drop.

05 Build a 12-month roadmap with quarterly milestones.

DELIVERABLES

- Maturity scorecard
- Roadmap
- Metric definitions

COMPLETION CHECK

Every roadmap item is linked to a scored gap and a metric that will show whether it worked.

LAB 32 VIDEO 32 60 MIN BUILDS ON: ALL

Plan your skills and next steps

Objective. Rate your skills against the whole course, choose a specialization, and write a plan that includes regular incident-simulation practice.

STEPS

01 Rate yourself 1–5 on each Part of the course.

02 Choose a specialization: forensics, malware analysis, threat hunting, incident command or program leadership.

03 Pick one certification that fits it.

04 Write a 90-day plan in three 30-day blocks.

05 Schedule a recurring simulated incident (for example monthly) and a public breach analysis (for example every two weeks).

DELIVERABLES

- Skills self-assessment
- 90-day plan
- Practice calendar

COMPLETION CHECK

The first practice session is on your calendar.